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RESULTS FROM THE ALS-U STORAGE RING ALIGNMENT SYSTEM PROTOTYPE*

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Abstract

ALS-U stability and alignment requirements coupled with tight space constraints present in the existing ALS prompted a new design for the storage ring support and alignment system. A prototype has been built and tested with alignment accuracy results in the 30 micron range and stability results in the 35 nm range. The new design overcomes distinct ergonomic challenges and reliability failures of earlier hardware iterations. The prototype has also been tested to an alignment time requirement needed to minimize dark time--the phase of the program where storage ring alignment occurs. This paper presents the innovative solutions implemented for the alignment system prototype to address the unique problems of ALS-U.

ALIGNMENT SYSTEM DESIGN

Background and Architecture

We previously tested the baseline design of the alignment system along with vibration stability, thermal performance and other subsystem prototype tests [1]. However, the baseline design experienced mechanical failures of roller bearings on two separate occasions so we built and tested a new alignment system. In the new design, the raft is supported on three sets of spherical bearings on Airloc load levelers in a symmetric configuration during alignment. The load levelers are mounted on a plinth which is bolted to an embedded plate in the ALS concrete floor. These load levelers are topped with custom 954-bronze (conforming to ASTM B505) bushings coated in Molykote G-0010 (a mineral-oil-based grease thickened by a di-urea system) to provide a low friction surface resistant to stick-slip behavior.

Once the raft is aligned, an M20 stud that penetrates through the center of each load leveler is torqued to 134 Nm for a preload of approximately 48 kN. For enhanced vibration stability, two additional load levelers are implemented with non-lubricated 954-bronze bushings. These have the additional benefit of providing a higher friction joint to lock the rafts in place for long-term positional stability. An overview of the alignment system design is shown in Figure 1 and the entire SR raft assembly prototype is shown in Figure 2. Figure 3 is a top view of the plinth showing details the bushing configuration.

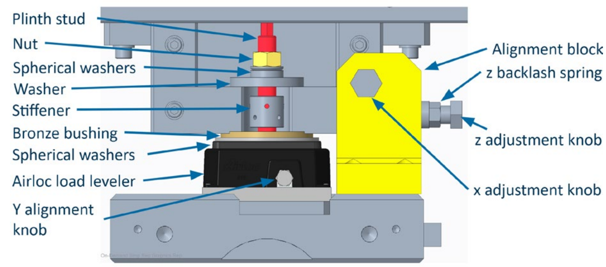

Figure 1: Alignment system CAD model.

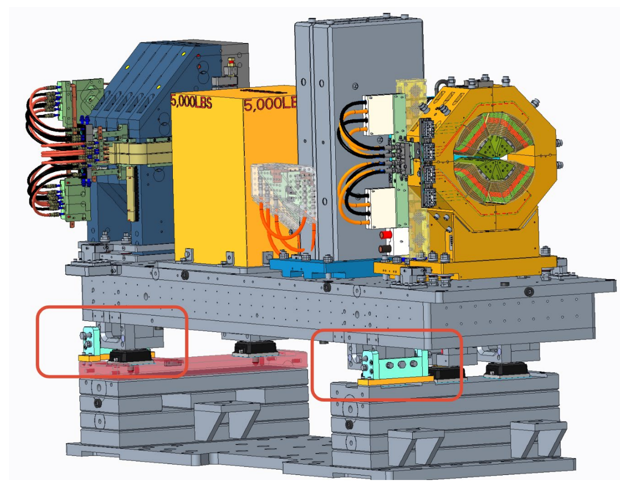

Figure 2: SR raft assembly prototype CAD model.

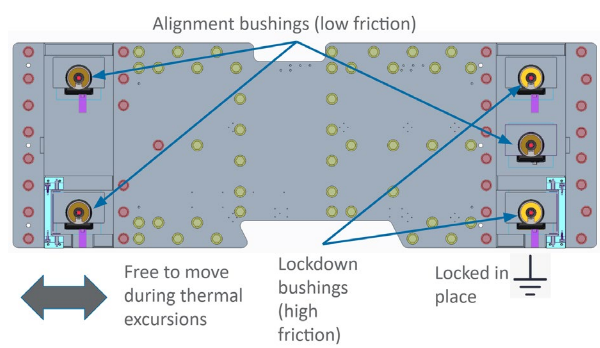

Figure 3: Bushing configuration.

Adjustment Mechanisms

Adjustments are made in 6 degrees of freedom (6-DOF) using 4 alignment screws in the x-direction, 2 alignment screws in the z-direction and 3 adjustable load levelers in the y-direction. Extra-fine M20 screws with a 1.5 mm pitch are used to adjust in the x- and z-directions, leading to 42 µm translation per 10 degree turn of the screw. The load leveler uses a sliding-wedge system to raise or tilt the raft with a 10 degree rotation of its adjustment screw, leading

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to 10 µm translation. The screws have custom 38 mm diameter rounded ends that push and slide against a Society of Automotive Engineers (SAE) 4140 steel (known as chromoly steel) plate for precise, repeatable control.

All x- and z-direction adjustment screws are located near the inboard side of the raft, allowing alignment from the aisle-side of the raft. This is an additional requirement on the system design since the outboard side of the rafts are often inaccessible, with radiation shielding and beamline hardware blocking access. The only adjustment mechanisms that are more difficult to reach are the load levelers on the outboard side, used for adjustment in the y-direction, and its associated nuts. Both are accessible using custom tooling from the inboard side.

Backlash springs apply force in the x- and z-direction for two purposes: allowing fine control by maintaining force on the leading edge of the adjustment screw, and providing a force great enough to push the raft towards the active adjustment screw so that adjustments can be made only from one side of the raft if desired. The spring assemblies are threaded on the outside so that their zero-force position can be set as desired, to allow a range of preloads. The combination of a "hard" adjuster (screw) and a "soft" adjuster (spring) is a versatile combination maximizing flexibility during fine alignment. A cross section of a backlash spring assembly is shown in Figure 4.

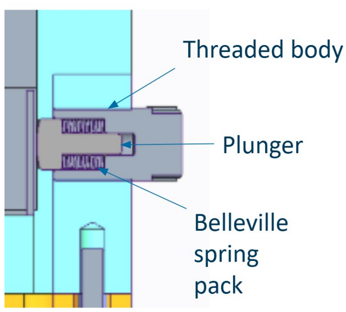

Figure 4: Backlash spring cross section view.

ALIGNMENT TEST

Measurement System

Two measurement systems are used to align the raft: a linear variable differential transformer (LVDT) network and a laser tracker pair. Instrumentation is shown in Figures 5 and 6. As shown in Table 1, both methods agree on raft position measurements to within 8 µm. The LVDT network has eight individual transducers and was used because of its lower instrumentation uncertainty compared to the laser tracker pair. Uncertainties of the two instrumentation approaches are shown in Table 2. Laser trackers are the baseline approach for raft to raft alignment in the ALS tunnel, however, an LVDT network for raft-to-raft alignment is under consideration since performance is similar but use in the tunnel may be quicker and provide less uncertainty in the raft position measurement.

Requirements and Results

The test addressed the key alignment requirements of the SR raft alignment system: alignment precision and time-to-align. Table 3 provides details of the portion of the total system alignment budget allocated to raft alignment.

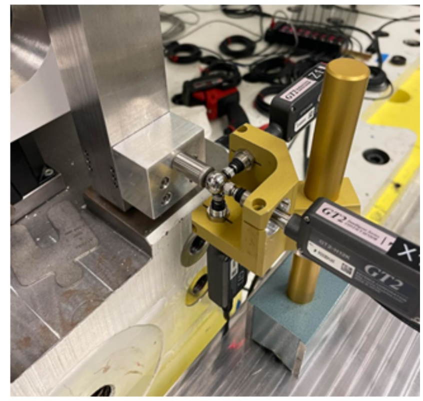

Figure 5: One of four LVDT assemblies.

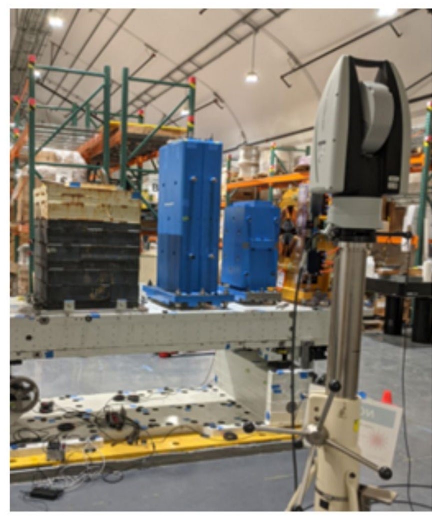

Figure 6: One of two laser trackers.

Table 1: Instrumentation Results Comparison

Test ID	X [µm]	Y [µm]	Z [µm]
Test 1	-3	-7	4
Test 2	8	-2	5
Test 3	5	-2	1
Test 4	-1	-2	-1
Test 5	0	5	-5

Table 2: Instrumentation Uncertainty.

Instrument	Uncertainty [µm]
LVDT Network	2 @ 1σ
Laser Tracker Pair	20 @ 1σ

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These results do not include instrumentation uncertainty since it is already accounted for in the system level alignment budget. The test also addressed time-to-align, an additional key requirement. The total time-to-align a raft from an arbitrary starting point to a predefined 6-DOF coordinate position must be less than 180 minutes to satisfy the top-level requirement that the entire storage ring can be re-aligned in a 3 week period [2].

Table 3: Key Requirements

Description	Requirement	Unit
x-direction	+/-30	µm
y-direction	+/-30	µm
z-direction	+/-120	µm
Rotation about x-axis (pitch)	+/-20	µrad
Rotation about y-axis (yaw)	+/-20	µrad
Rotation about z-axis (roll)	+/-120	µrad

We performed ten run-for-record tests. Alignment requirements were met in every test, as shown in Table 4. Three bushing types were tested: graphite-impregnated sintered bronze, oil-impregnated sintered bronze, and 954-bronze coated in Molykote G-0010.

Table 4: Bushing Types and Alignment Times

Test ID	Coating	Time [minutes]
Test 1	Graphite	75
Test 2	Graphite	60
Test 3	Graphite	75
Test 4	Graphite	75
Test 5	Oil	50
Test 6	Oil	45
Test 7	Oil	45
Test 8	Oil	45
Test 9	Molykote	30
Test 10	Molykote	30

The new alignment system was used to move the raft in all 6 DOF by a predefined amount at each step (Test ID). Translation step sizes were between 0.020 mm and 5.500 mm while rotational step sizes were between 20 µrad and 200 µrad. Figures 7 and 8 present the key results from the alignment test. The red dotted lines identify the requirement limits for x-, y-, pitch, and yaw directions while the blue phantom lines identify requirement limits for z- and roll directions. Results were consistent from test to test.

One key finding was that larger moves required more time—up to 45 additional minutes. Therefore it will be important to rough align the rafts before installation in the ALS so that less time will be spent on alignment in the tunnel. Additionally, due to instrumentation drift, larger moves will be subject to larger positional errors.

Another key finding was that while errors can be minimized to <5 µm, it takes longer than the 180 minute requirements to achieve. For x- and y-directions, the optimum strategy for ALS alignment, it appears, is to achieve <15 µm, then move to the next raft to avoid spending excessive time aligning the storage ring.

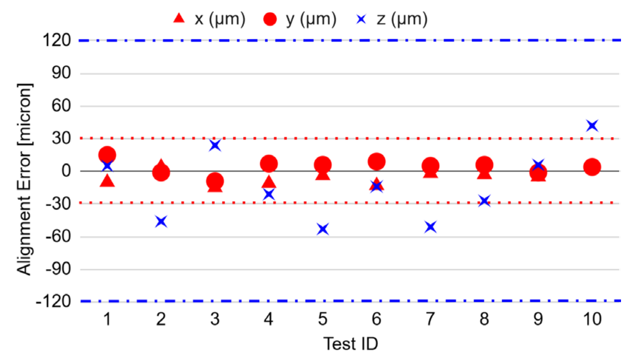

Figure 7: Alignment results – axes of translation.

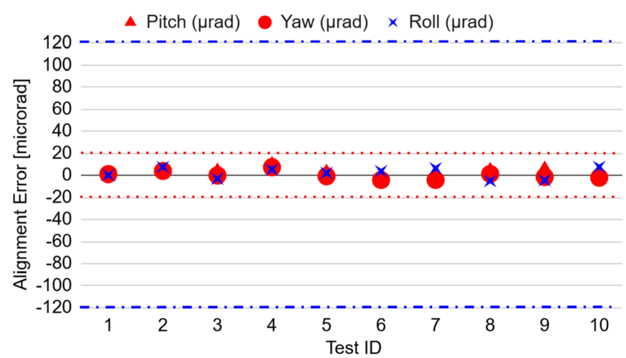

Figure 8: Alignment results – axes of rotation.

In addition, Molykote G-0010 bushings were the quickest to align and had the least pronounced stick-slip behavior, while graphite bushings were the slowest to align and had the most pronounced stick-slip behavior. In addition, Molykote G-0010 is a standard lubricant already qualified for use in the ALS.

CONCLUSION

The new SR raft alignment system meets all position and time-to-align requirements using 954-Bronze bushings coated in Molykote G-0010 which out-performed the alternative bushings tested. Unlike the original alignment system design, no hardware failures interrupted this test which can be taken to mean the new design is more reliable. Additionally, all adjustment mechanisms are accessible from the inboard side of the raft assembly. The as-tested LVDT network performed similarly to the laser tracker pair and so will be developed further for use in the ALS tunnel.

REFERENCES

[1] C. Steier *et al.*, "Design and construction progress of ALS-U", in *Proc. IPAC'24*, Nashville, TN, USA, May 2024, pp. 1313-1316. doi:10.18429/JACoW-IPAC2024-TUPG38

[2] R. Miller, "SR Prototype Raft Alignment Test Plan", Lawrence Berkeley National Laboratory, Berkeley, CA, US, internal report, Jan 2025.